科学真实性、准确性与完整性内部审查文档

对应稿件： *Limited concordance between storm-tide magnitude and standardized connected-lowland screening across Northwest Europe*

审查定位： code-data-table-figure-manuscript 穿透式内部一致性审计，不冒充独立同行评议

自动约束状态： 全部通过

1. 审查结论

本轮首先修复会改变空间推断的实现问题，而不是先润色结论。旧 `coastal_sector()` 以经纬度阈值给站点分区，曾把 Fishguard、Holyhead、Cuxhaven 和 Tregde 等站放入不合理海区；它已从统计路径移除。现在 74 个候选站通过 `config/coastal_spatial_blocks_v2.csv` 显式连接到 coastal basin 以及 100、150、250 km 三套版本化空间块。块内 restricted permutation 的双侧概率也改为经验双尾，不再假定受限零分布以 0 为中心。

当前可确认的是：68 站主表、2,960 条地形稳健性记录、三套空间统计、图件和文稿由同一 source-table 体系生成；面积、拓扑和指标分解契约通过。当前证据支持的是 limited rank concordance 和 decision-level disagreement。它不支持“显著负相关定律”“错配异常大于空间随机”“connected terrain 是真实洪水排序”或“二指标组合已经提高预测能力”。

2. P0 问题及处理证据

1. **空间错分：已修复并重算。** 经纬度 `if` 规则不再参与推断。地图、块内置换、分块调整关联、块重采样、leave-one-block-out 和 Top-k null 统一读取版本化配置；已知错误映射由自动测试固定。

2. **受限置换双侧概率：已修复并重算。** 当前概率是 `2 * min(P(null <= observed), P(null >= observed))` 的加一校正经验双尾值，允许空间受限零分布偏离 0。

3. **复现入口：已分层。** `python scripts/reproduce.py --mode derived` 从带 SHA-256 的紧凑派生输入重建论文包；`--mode full` 从本地上游原始产品开始。README 不再把 derived rebuild 描述成 raw-to-paper。

4. **审稿包依赖闭包：已重构并设独立门禁。** `src/`、全部 Python 脚本、测试、配置及 figure-source tables 由目录规则纳入，不再依赖易漏项的脚本白名单；最终 ZIP 在包外以临时目录、全新虚拟环境和九个确定性输出哈希独立验收。

5. **运行环境：已补齐。** `pyproject.toml` 声明 xarray、netCDF4、rasterio、pyproj、SciencePlots 与 pytest 等实际依赖；`environment-lock.yml` 记录本轮平台版本。系统全局 pytest 插件可能冲突，因此正式门禁固定 `PYTEST_DISABLE_PLUGIN_AUTOLOAD=1`。

3. 数据产品与 mask 审计

product	release	tile_tag	vertical_reference	release_cap_m	paper_v1_0_cap_m	mask_outside_support_fill	official_255_cells_in_primary_analysis_windows	outside_support_cells_in_primary_analysis_windows	note
DeltaDTM	v1.1.1	DeltaDTM v1.1.1 / DeltaDTM.jl v1.1.1	EGM2008	30	10	254	0	3320	The 2024 paper documents v1.0; the downloaded v1.1 README documents the 30 m EGM2008 cap. Outside-support fill is kept distinct from class 255 and excluded from all land denominators.

主流程只允许官方 class 0 land 进入 represented-land denominator；class 1 ocean 是基础 marine seed，class 3 river 仅在与 ocean component 连通时参与水体种子，class 2 lake、class 255 和窗口外 class 254 均排除。这里的 `connected` 是 raster component 分类，不是水动力连通、洪水历时或防洪工程失效判断。

4. 自动科学约束

Check	Pass	Observed	Criterion or meaning
Robustness-grid rows	True	2960	74 stations x 40 configurations
Primary stations	True	68	68 unique matched stations
Explicit spatial-block coverage	True	74	74 configured candidates, no missing block
Known coastline corrections	True	True	Fishguard/Holyhead/Cuxhaven/Tregde reviewed mappings
Connected area <= below-threshold area	True	0	violations
Four-neighbour <= eight-neighbour	True	0	violations under identical configuration
Metric decomposition identity	True	1.13687e-12	maximum absolute percentage-point error
Official class 255 in primary windows	True	0	cells
Outside-support sentinel distinct	True	3320	class-254 cells, excluded
Threshold grid complete	True	884	68 stations x 13 thresholds from 0 to 3 m
Water/land topology grid complete	True	272	68 stations x four explicit topology combinations
Coastline-window sensitivity complete	True	68	stations
Decision audit complete	True	68	stations
Coordinate match review complete	True	68	68 included coordinate-plausible pairs; six distance-excluded candidates are not counted

这些检查能证明代码输出满足自身声明的集合、面积与拓扑关系，也能阻止已知回归。它们不能证明输入 DEM 没有系统误差，不能替代共同垂向基准，更不等同于独立洪水观测验证。

5. 空间统计证据边界

terrain_metric_label	block_scale_km	n_spatial_blocks	spearman	kendall	sector_adjusted_rank_association	sector_stratified_permutation_two_sided_p	top20_n	top20_observed_overlap	sector_stratified_top20_null_q025	sector_stratified_top20_null_q975	sector_stratified_top20_lower_tail_p	spearman_sector_bootstrap_p025	spearman_sector_bootstrap_p975
lowland_prevalence	100	42	-0.309272	-0.187884	-0.395706	0.181582	14	1	1	1	1	-0.549305	0.033286
lowland_prevalence	150	34	-0.309272	-0.187884	-0.477079	0.0219978	14	1	1	2	0.332667	-0.548781	-0.0072409
lowland_prevalence	250	25	-0.309272	-0.187884	-0.41672	0.020198	14	1	1	2	0.331967	-0.582751	0.0704137
conditional_connectivity	100	42	0.104533	0.06924	0.230842	0.490151	14	5	4	5	1	-0.161064	0.330231
conditional_connectivity	150	34	0.104533	0.06924	0.138767	0.584742	14	5	3	6	0.917308	-0.161052	0.315844
conditional_connectivity	250	25	0.104533	0.06924	0.147126	0.464154	14	5	2	6	0.971203	-0.150571	0.326691
connected_lowland_share	100	42	-0.333168	-0.210711	-0.349591	0.135186	14	1	1	1	1	-0.550994	-0.0272113
connected_lowland_share	150	34	-0.333168	-0.210711	-0.455223	0.0251975	14	1	1	2	0.332467	-0.567171	-0.0273174
connected_lowland_share	250	25	-0.333168	-0.210711	-0.41179	0.0119988	14	1	1	4	0.0985901	-0.578765	0.0163185
connected_area	100	42	-0.340955	-0.208077	-0.401135	0.121988	14	1	1	1	1	-0.565968	-0.0190467
connected_area	150	34	-0.340955	-0.208077	-0.481675	0.0319968	14	1	1	2	0.331567	-0.558951	-0.0456077
connected_area	250	25	-0.340955	-0.208077	-0.433954	0.0129987	14	1	1	4	0.0977902	-0.576907	0.0158461

100、150 和 250 km 方案分别包含 42、34 和 25 个近似沿岸连续块。RP10 与 connected-land share 的原始 Spearman rho 为 -0.333；块内经验双侧 p 随分块尺度为 0.135、0.025 和 0.012，而保持地理组成的块重采样范围为 [-0.551,-0.027]、[-0.567,-0.027] 和 [-0.579,0.016]。两类结果不矛盾：前者问同一组空间块内随机重排后是否常见，后者问换一组被抽到的地理块时估计是否稳定。250 km 范围触及 0，因此稿件不建立空间稳定的负相关定律。

Top-k 空间零模型

block_scale_km	n_spatial_blocks	top_n	observed_overlap	sector_stratified_null_q025	sector_stratified_null_q975	sector_stratified_lower_tail_p
100	42	14	1	1	1	1
150	34	14	1	1	2	0.326667
250	25	14	1	1	4	0.0982902

Top-20% 两表各 14 站，实际只重合 1 站，但该值在三套空间零模型包络内。1/14 是筛查决策的描述性恢复度，不是“低于随机”的显著性证据。

6. 指标构念与决策稳定性

connected-land share 被显式拆为 lowland prevalence 乘以 conditional connectivity。主结果中前者与 storm tide 的 rho 为 -0.309，后者为 +0.105；所以低 composite 值不能一概解释为“海洋连通差”。

Decision diagnostic	Observed
Median absolute rank displacement	26
Maximum absolute rank displacement	64
Always terrain Top-20% across eligible configurations	6
Never terrain Top-20% across eligible configurations	40
Pareto-nondominated sites	5

绝对排名位移中位数为 26。地形 Top-20% 选择频率来自 40 个预设配置中的可用子集，它度量方法选择下的清单稳定性，不是洪水概率。四象限和 Pareto 结果用于说明 costly second-stage modelling 的候选清单如何变化，不把 terrain ranking 当作真值。

水体/陆地邻域、阈值与窗口

water_neighbours	land_neighbours	terrain_metric	n_sites	water_vs_terrain_spearman	terrain_rank_spearman_vs_primary_8water_4land	top20_n	top20_overlap_with_primary	top20_membership_changes
4	4	connected_reference_land_pct	68	-0.427492	0.862007	14	12	4
4	4	connected_area_km2	68	-0.430927	0.862122	14	12	4
4	4	conditional_connectivity_pct	68	0.0888363	0.957201	14	14	0
4	8	connected_reference_land_pct	68	-0.365958	0.88785	14	11	6
4	8	connected_area_km2	68	-0.371493	0.896362	14	11	6
4	8	conditional_connectivity_pct	68	0.012005	0.867483	14	13	2
8	4	connected_reference_land_pct	68	-0.333168	1	14	14	0
8	4	connected_area_km2	68	-0.340955	1	14	14	0
8	4	conditional_connectivity_pct	68	0.104533	1	14	14	0
8	8	connected_reference_land_pct	68	-0.319846	0.961102	14	11	6
8	8	connected_area_km2	68	-0.329427	0.966828	14	11	6
8	8	conditional_connectivity_pct	68	-0.00615544	0.900571	14	13	2

terrain_metric	n_sites	spearman	kendall	interpretation
connected_share_auc_0_3m_pct	68	-0.298202	-0.187006	threshold-integrated standardized EGM2008 terrain screen; not a flood probability
connected_area_auc_0_3m_km2	68	-0.31328	-0.186128	threshold-integrated standardized EGM2008 terrain screen; not a flood probability
below_share_auc_0_3m_pct	68	-0.289232	-0.175593	threshold-integrated standardized EGM2008 terrain screen; not a flood probability

terrain_metric	n_sites	terrain_rank_spearman_oriented_vs_square	water_vs_oriented_spearman	water_vs_square_spearman	top20_n	top20_overlap_oriented_vs_square	top20_membership_changes	interpretation
connected_reference_land_pct	68	0.803794	-0.247662	-0.333168	14	11	6	mask-derived window-geometry sensitivity, not an alternative flood model
connected_area_km2	68	0.741535	-0.254151	-0.340955	14	11	6	mask-derived window-geometry sensitivity, not an alternative flood model
below_reference_land_pct	68	0.892812	-0.201779	-0.309272	14	14	0	mask-derived window-geometry sensitivity, not an alternative flood model

主流程是 water-8 / land-4。严格 water-4 / land-4 与主流程地形排名 rho 为 0.86、Top-20% 改变 4 个成员。0-3 m 阈值积分和近似沿岸窗口仍给出有限一致性，但沿岸窗口与方窗仅重合 11/14 个 terrain Top-20% 成员。这些结果支持区域结论的方向，也明确暴露单站优先级的方法依赖性。

7. COAST-RP 匹配审计

68 个纳入站点最大 great-circle 匹配距离为 2.811 km；以下 6 站因超过 6 km 预设规则被排除：

station	recomputed_match_dist_km	coast_rp_source_lon	coast_rp_source_lat	match_exclusion_reason
Cordemais France	22.6474	-2.19	47.278	excluded_nearest_coast_rp_point_gt6km
Le Pellerin France	28.388	-2.014	47.014	excluded_nearest_coast_rp_point_gt6km
Nantes Usine Brulee	39.423	-2.014	47.014	excluded_nearest_coast_rp_point_gt6km
Donges France	8.27799	-2.19	47.278	excluded_nearest_coast_rp_point_gt6km
Montoir De Bretagne	6.89698	-2.19	47.278	excluded_nearest_coast_rp_point_gt6km
Paimboeuf France	14.3371	-2.19	47.278	excluded_nearest_coast_rp_point_gt6km

坐标级审阅对 74 个纳入匹配全部给出“地理位置可接受”，其中 11 个标记为河口或半封闭海湾。该审阅没有 coastline component 或沿岸路径距离，字段明确写为 `opposite_bank_flag = not_determined_from_coordinate_review`，不能升级为 hydrodynamic equivalence。

8. national/local DTM product-and-datum cross-check

station	delta_connected_2m_reference_land_pct	local_connected_2m_reference_land_pct	delta_connected_2m_area_km2	local_connected_2m_area_km2	connected_pixel_iou_after_reprojection	local_dtm_vertical_datum_note
Sheerness	42.7002	9.62959	20.3643	4.61207	0.214623	Ordnance Datum Newlyn / EA product native datum
Newlyn	0.44611	0.416648	0.276293	0.259144	0.527548	Ordnance Datum Newlyn / EA product native datum
Lowestoft P024 Uk	1.41616	6.16525	0.669507	2.92848	0.110977	Ordnance Datum Newlyn / EA product native datum
Immingham P026 Uk	0.405551	0.078999	0.24946	0.0487939	0.0417246	Ordnance Datum Newlyn / EA product native datum
Denhelder Hel Nl	71.7296	4.36493	20.8545	1.27433	0.0554507	Normaal Amsterdams Peil / AHN product native datum
Delfzijl Del Nl	96.7041	89.7843	61.3031	57.1869	0.925796	Normaal Amsterdams Peil / AHN product native datum
Hoek van Holland	2.63966	1.43861	1.41098	0.771845	0.217064	Normaal Amsterdams Peil / AHN product native datum

七站对照记录共同 target grid、transform、mask resampling 和文件 SHA-256，但 DeltaDTM、Ordnance Datum Newlyn 与 Normaal Amsterdams Peil 未统一。因此差异同时包含高程产品、垂向基准、水平分辨率和 provider WCS 重采样；本文只称 diagnostic product-and-datum sensitivity，不称 validation，也不把七站变化比例外推到 68 站。

9. 图件来源与视觉检查

Figure	Pixels	Embedded DPI	Source data	Visual/data QA
Fig1_process_terrain_design.png	3542 x 2404	500	Primary_station_spatial_blocks.csv; Primary_sample_flow.csv; coastal_spatial_blocks_v2.csv	original-resolution inspection required after each final rebuild; no cell enlargement
Fig2_regional_screening_agreement.png	3719 x 3362	500	Primary_terrain_components_68stations.csv; Return_period_sensitivity.csv; Spatial_block_bootstrap_intervals.csv; Topk_spatial_null_curves.csv; Decision_instability_station_metrics.csv	original-resolution inspection required after each final rebuild; no cell enlargement
Fig3_connected_terrain_contrasts.png	3687 x 3934	500	Fig3_predefined_site_selection.csv; Fig3_threshold_response_source.csv; frozen DeltaDTM windows	original-resolution inspection required after each final rebuild; no cell enlargement
Fig4_metric_decomposition_robustness.png	3816 x 3320	500	Primary_terrain_components_68stations.csv; Denominator_adjusted_associations.csv; Water_land_topology_summary.csv; Rank_robustness_one_at_a_time.csv; Threshold_integrated_associations.csv; Coastline_window_sensitivity_summary.csv	original-resolution inspection required after each final rebuild; no cell enlargement
FigS1_water_level_definition.png	3444 x 1458	500	Fig2_water_level_divergence_with_uncertainty.csv; source-verified tidal metadata	original-resolution inspection required after each final rebuild; no cell enlargement
FigS2_process_coherence.png	5122 x 2354	500	Fig3_driver_correlations.csv; event-composite source tables	original-resolution inspection required after each final rebuild; no cell enlargement
FigS3_connectivity_gallery.png	3462 x 3415	500	Primary_terrain_components_68stations.csv; frozen DeltaDTM windows	original-resolution inspection required after each final rebuild; no cell enlargement
FigS4_elevation_product_datum_sensitivity.png	3591 x 2975	500	TableS_local_dtm_product_datum_crosscheck.csv	original-resolution inspection required after each final rebuild; no cell enlargement
FigS5_coastrp_match_audit.png	3466 x 1591	500	Match_distance_audit_74stations.csv; COAST_RP_same_coast_match_audit.csv	original-resolution inspection required after each final rebuild; no cell enlargement

Figure 3 的三站由机器可读规则选出。低值站 inset 只放大原像元位置，不增加 connected 面积；9 子图分别展示分类、指标分解和实算阈值曲线。每次最终重建后仍须以原始分辨率检查标签遮

挡、图例、字体和颜色，并把像素层面的视觉判断与 source-table 数值核对分开记录。

10. 关键文件 SHA-256

File	Bytes	SHA-256
data/raw/coast_rp/COAST-RP.zip	6798999	13242fe8b6822304203 57dbac202b10590f41dc0c9e378b9c2c69f91962704c9
data/raw/gssr/metadata/eraint.geojson	307194	aab77394a0bf87b3ae611131e26b489a7abe9dd67add071aeee0da4b3bab46e6
data/raw/deltadtm/index/README.md	2326	9da10b0576ff4368a756bd5646b091fab6b7c7c7d217a6f758fbe0803f826e00
data/raw/deltadtm/zips/mask_tiles.zip	428178062	3f8c6999f90346359e60cc6c757786d43fd1cb65e4ab13125faaa80606487775
config/coastal_spatial_blocks_v2.csv	6730	05df4af816e3287c6fbf9d7fe9c8ee8ad1b66f3ef2d9533f626e2d008d4e8791
data/upstream_raw_manifest.csv	77446	c3567da4627f59e3bb00f40f6957db0770dac691e4c2a01acb8b2ef73f57c620
data/derived_input_manifest.json	3004	9dfa418329cfabec84fe0e2f5abbbeca17bdf05ff8b542a86369da9b6d639154
data/figure_source/Primary_terrain_components_68stations.csv	35122	f1e5eef717a36ee71ec82bbef8da9f487a1ae843c9e1cc6e9e3e791bf28c58f2
data/figure_source/Terrain_robustness_grid_74stations.csv	1481287	6142194e3fe7354c8c5cdf48d12c5d95a2d40364b4d97138947d43666b115661
data/figure_source/Return_period_sensitivity.csv	25437	1b5517fc32b1a73413764b647c1449e830d6e7639dd34bba6ff9387d4049f11a
data/figure_source/Spatial_block_bootstrap_intervals.csv	6972	6883bc444e5e34edc53325889fd1e57ebc93e0fbee67a06f723d9b99b034b814
data/figure_source/Topk_spatial_null_curves.csv	19132	24ca73f5eca3e5f25318afd65cda2bca53702801e32806d0c5f07ddee16dd8f9
data/figure_source/Water_land_topology_sensitivity_68stations.csv	72596	a763ad91307e56be4ca11cb538724bc1d78b7b5d022685922cfd879a0e8a53c8
data/figure_source/Threshold_response_0_3m_68stations.csv	225103	274719124c9f803ae29315d4adba1d7eda94c39aeda88d306b8a9a15b6c47abf
data/figure_source/Coastline_oriented_window_68stations.csv	27461	57330dabb4f6bbea9af86c2c484dee117f9bce5d3abea5fa9f7346e3af42ff8c
data/figure_source/Decision_instability_station_metrics.csv	40312	427dfdb53466e679b0a89be80bd12dea34d7fead7927ca8efe61e031693e6a41
data/figure_source/COAST_RP_same_coast_match_audit.csv	66453	ce2ee8fe792f566cdb1f5203aff73b4ddf1131879096ab3578da6f5260794970
data/figure_source/TableS_local_dtm_product_datum_crosscheck.csv	9735	20a1346c2a15016de6d8f6cc983c296e8728a2a452acb24750df6da7ebb6b597
scripts/reproduce.py	8137	ac6df5d8b9aeaf2c314407581e0a3aa83cced98cb8b3da797b7593fb1100ae96
scripts/rebuild_frozen_primary_analysis.py	25305	9c306499e07e25a99cc5b1f75221300ebed755f4ad5a477d9063d517323736c1
scripts/add_external_review_diagnostics.py	31815	22e869b23a08ca1432e940d036133b7d2422f025db764a6a684f06adb97b57c6
scripts/make_major_revision_figures.py	46635	71e9f9484a04d29b7f1544270f67aa73a5ec2e7371771ac3c851d84574ca04f6
tests/test_scientific_contracts.py	14920	7161d27bf4c9710186ac789fc4abecb707e209c20ea5e80c07e6153dd91294c1

File	Bytes	SHA-256
src/coastal_screening/__init__.py	332	4ed9dcca6879041b6fb6252d7760c66b7d89ee23e33242eb4bb113b36e8f3500
src/coastal_screening/connectivity.py	1330	9a1eedcdd0aefa3659a25f3df7f0eb6404d2c24ed45855ac71246f4478962199
src/coastal_screening/spatial_blocks.py	3973	bc0c1f953c05f6f7b4ce74bb498a9ecc49649468804a1dbc33a5c3a91876ade91
src/coastal_screening/statistics.py	3122	035d10af40bcf808836d47fba87ab401e0c6691adfa94c84b486f2e61977a2b3
data/raw/local_dtm/delfzijl-del-nl_local_dtm_10m.tif	2204083	a8b3589a24ca9dbcb00d18e24c916e193d37a9b73034c185bf29fcf80a9cbb35
data/raw/local_dtm/denhelder-hel-nl_local_dtm_10m.tif	1310648	b950e177da667509c85bb8a4dc77c4dcbac5b56d85507ea706e968568af15e79
data/raw/local_dtm/hoekvanholla-hvh-nl_local_dtm_10m.tif	1848802	e960ac322ae8e703055a75739d6dee6a65c2ee367e9dc0dd8af1c7f6ff600563
data/raw/local_dtm/immingham-p026-uk_local_dtm_10m.tif	9437675	8567cb377b27480b51f3dbee6ff5501784f3eed237e0ec7e8058ff09789ed6e7
data/raw/local_dtm/lowestoft-p024-uk_local_dtm_10m.tif	6291923	d95f97c91d4df295e6113b98fccbdab4d3bedac1a35c3751464e0b8e61ab2819
data/raw/local_dtm/newlyn-p001-uk_local_dtm_10m.tif	9437675	fef345c91868dd64e5a2e11180ad066b7a1f8ce92305cf26990d70fd0c9e51c2
data/raw/local_dtm/sheerness-p015-uk_local_dtm_10m.tif	9437675	40c81f2a969e0044184d5fb40ebc11728dca5376d2707386e29c90def027cadd

大体积上游数据不能凭文件名认定相同。full 模式在计算前核验 `data/upstream_raw_manifest.csv`；derived 模式核验 `data/derived_input_manifest.json`。本表用于确认论文、报告和发布包是否来自同一输入与代码基线。

11. 尚未消除的风险

Risk	Status	Evidence boundary
垂向基准统一	未完成	DeltaDTM/ODN/NAP 尚未转换到共同 local-MSL reference；S4 只能解释为产品、基准、分辨率和服务重采样的联合敏感性。
独立真实洪水基准	未完成	检索到的 2026 European total-water-level storms 数据仍是水位过程，不是站点级动态淹没真值；没有用它伪装外部验证。
空间块定义	部分完成	74 站已由版本化表显式审阅并修正已知错分；100/150/250 km 标签是近似沿岸连续分组，不是精确测得的 shoreline distance。
COAST-RP 同岸环境	部分完成	坐标和地图审阅通过 68 对匹配并标记 11 个河口/复杂岸线点；这不证明同一沿岸水动力环境，仍需 coastline component/alongshore-distance 数据。
独立站点框架	受限	68 站来自 GSSR metadata inventory，且英国与爱尔兰占 45 站，不能外推为均匀欧洲海岸样本。
公开复现	计算链已通过	full 模式已核验 278 个上游文件并完成 raw-to-paper 重建；最终审稿 ZIP 的 clean-room 结果由包外 reviewer_archive_validation_receipt.json 独立记录。
作者元数据与 DOI	待补充	authors、affiliations、funding、contributions 及正式归档 DOI 尚不能由代码推断。

12. 可验收的后续目标

1. **共同垂向基准：** 至少 3 个英国和 3 个荷兰 case 转换到明确共同 coastal-MSL reference，记录 offset 来源、公式、不确定性和重算结果；否则 S4 永久保持 sensitivity 定位。

2. **独立 outcome：** 只在存在站点可匹配的 observed 或 dynamic flood-hazard outcome 后，预先定义并比较 water-only、terrain-only、combined screen；完成前不使用 predictive complementarity。

3. **独立海岸样本：** 用不依赖 GSSR inventory 的 station/shoreline inventory 重算，公开全部候选、排除原因和沿岸采样覆盖。

4. **同岸匹配：** 计算 coastline component 与 alongshore distance，逐站固定 opposite-bank/estuary 判定；当前坐标审阅不得替代这一工作。

5. **发布复现：** 在全新临时目录解压 reviewer ZIP，安装项目、运行 derived reproduction、测试并比较确定性科学输出哈希；记录 release tag、commit 与 ZIP SHA-256。

6. **投稿元数据：** 补齐 authors、affiliations、funding、contributions 和正式归档 DOI；HTML、Markdown、PDF 中不留“待补充”。

13. 复现入口

- 审稿人快速复现：`python scripts/reproduce.py --mode derived`
- 上游文件预检：`python scripts/reproduce.py --mode full --verify-only`
- 完整 raw-to-paper：`python scripts/reproduce.py --mode full`
- 科学合同：`PYTEST_DISABLE_PLUGIN_AUTOLOAD=1 python -m pytest tests -q`
- 发布门禁：`python scripts/run_major_revision_rebuild.py`